

Final Project

[Start Assignment](#)

- Due Tuesday by 11:59pm
- Points 100
- Submitting a text entry box or a file upload
- File Types rmd, pdf, and html
- Available after May 4 at 12am

This is an individual project. Collaboration with others is strictly prohibited. Please submit a signed Honor Code statement along with your project submission. [Honor code-2026_DSII.pdf](#)
(<https://courseworks2.columbia.edu/courses/240438/files/27446183?wrap=1>)

Background

A research study aims to identify key factors that predict the severity of COVID-19 illness. This study collects demographic information, clinical variables, and disease severity among participants infected with COVID-19 between 2021 and 2023. The goal is to develop a robust prediction model for COVID-19 severity and understand how predictors impact the risk of severe infection.

The researchers have expressed interest in exploring both advanced predictive modeling techniques, specifically boosting and support vector machines (SVM), and simpler, more interpretable methods.

Question

The researchers are specifically interested in the following questions:

1. Do boosting and SVM provide superior predictive performance compared to simpler models?
2. Can we develop a predictive risk score (i.e., the predicted probability) that quantifies the chance of experiencing severe infection based on individual participant characteristics?
3. What are the key demographic and clinical factors that predict the risk of severe infection, and how do these factors influence the risk?

Report Components

Please complete a report that helps the researchers answer the questions of interest stated above. At a minimum, your report should include the following components:

- **Exploratory analysis.** Use summary statistics and appropriate visualization techniques to

explore the data. If you generate multiple figures, combine them into one figure with subfigures. Limit summary statistics to no more than one table.

- **Model training.** Provide a detailed description of the model training process. Your description should include sufficient detail so that another statistician can reproduce the same model(s). Include cross-validation plots (if there are multiple figures, combine them into one with subfigures) or tables to demonstrate the validity of the model training process.
- **Results.** Provide clear answers to each of the researchers' questions of interest. Provide a thorough interpretation of your model and assess its predictive performance.

Dataset

The full dataset (severity.RData

(<https://courseworks2.columbia.edu/courses/240438/files/27446165?wrap=1>) 

(https://courseworks2.columbia.edu/courses/240438/files/27446165/download?download_frd=1))

contains 10,000 participants and 15 variables, including demographic information, clinical measurements, and the disease severity outcome. The dataset is loaded into R as a data frame named `dat`.

Sampling Procedure

Create your own dataset by **randomly sampling 1,000 participants** from the full dataset, using **the last four digits of your UNI** as the random seed. Then split this dataset into a **training set (80%, $n = 800$)** and a **test set (20%, $n = 200$)**.

Example R code (using seed `xxxx` as a placeholder — replace with the last four digits of your UNI):

```
# Load the full dataset (loads object `dat` into your environment)
```

```
load("severity.RData")
```

```
# Set the random seed to the last 4 digits of YOUR UNI
```

```
set.seed(xxxx)
```

```
# Randomly sample 1,000 participants for your analysis
```

```
my_dat <- dat[sample(nrow(dat), 1000), ]
```

Codebook

A description of each variable is provided in the table below.

Variable Name (Column Name)	Description
ID (id)	Participant ID
Age (age)	Age
Gender (gender)	1 = Male, 0 = Female
Race and ethnicity (race)	1 = White, 2 = Asian, 3 = Black, 4 = Hispanic
Smoking (smoking)	Smoking status; 0 = Never smoked, 1 = Former smoker, 2 = Current smoker
Height (height)	Height (in centimeters)
Weight (weight)	Weight (in kilograms)
BMI (bmi)	Body Mass Index; BMI = weight (in kilograms) / height (in meters) squared
Hypertension (hypertension)	0 = No, 1 = Yes
Diabetes (diabetes)	0 = No, 1 = Yes
Systolic blood pressure (SBP)	Systolic blood pressure (in mm/Hg)

LDL cholesterol (LDL)	LDL (low-density lipoprotein) cholesterol (in mg/dL)
Vaccination status at the time of infection (vaccine)	0 = Not vaccinated, 1 = Vaccinated
Depression score (depression)	Higher scores indicate higher risk for depression
Severity of COVID-19 infection (severity)	Response variable; 0 = Not severe, 1 = Severe

Submission

Your submission should include two components:

1. **Analysis Report (PDF).** Follow the formatting requirements below.
 - **Font:** Times New Roman, 12 pt.
 - **Line spacing:** 1.5.
 - **Page limit:** The main text of your report should not exceed 4 pages. This limit applies to the text only; figures and tables do not count toward the page limit.
 - **Figures and tables:** All figures and tables should be placed in an Appendix at the end of the report, after the main text. Each figure and table must have a clear caption. Reference all figures and tables in the main text by their labels (e.g., “see Figure 1,” “as shown in Table 2”).
 - **No R code in the report.** Avoid inserting raw outputs from R functions; instead, demonstrate your ability to interpret and distill useful information from output tables.
 - Points will be deducted if the formatting requirements above are not followed.
2. **R code and outputs (Rmd + knitted file).** Your submission should contain all the code for conducting your analysis, as well as for the creation of any tables, figures, and numbers presented in your report.